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POSITION DETECTION DEVICE AND DETECTION DEVICE

Abstract

A position detection device for detecting the position of a rack shaft that steers the steering wheel of a vehicle by moving in the axial direction, includes a cylindrical rack housing that houses the rack shaft, a detector that detects the position of the rack shaft relative to the rack housing, a support member that supports the detector relative to the rack housing, and a seal member attached to the support member, wherein the rack housing has a through-hole that penetrates between the inner and outer circumference surfaces along the radial direction of the rack shaft. The support member has a support portion that is positioned in the through-hole to support the detector and a fixing portion that is fixed to the rack housing, and the seal member is in elastic contact with the outer surface of the support portion and the inner surface of the through-hole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present patent application claims the priority of Japanese patent application No. 2024-34641 filed on Mar. 7, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] This invention relates to a position detection device for detecting the position of a rack shaft of a vehicle and a detection device for detecting a physical quantity.

BACKGROUND OF THE INVENTION

[0003] As a position detection device for detecting the position of a rack shaft of a vehicle, the applicant of the present application has proposed the one described in Patent Literature 1. The position detection device described in Patent Literature 1 has a target made of high permeability material or high conductivity material fixed to the rack shaft, an excitation coil that generates an alternating magnetic field, and a detection coil with which the magnetic flux of the alternating magnetic field generated by the excitation coil chains. The excitation coil and detection coil are formed on a single substrate and extend along the axial direction of the rack shaft. The substrate is fixed to the lid of the rack housing, which has a metal body and a plastic lid. When the position of the target relative to the substrate changes as the rack shaft moves, the intensity of the magnetic field detected by the detection coil changes. The substrate corresponds to a detector of the position detection device described in Patent Literature 1.

[0004] Also, Patent Literature 2 describes the provision of a breathing valve in the housing of a vehicle steering device to mitigate internal pressure increases. The breathing valve consists of a membrane that allows air and water vapor to pass through, but blocks water droplets.

PRIOR ART LITERATURE

[0005] Citation List Patent Literature 1: JP 2023-174533 [0006] Patent Literature 2: JP 2019-044881

SUMMARY OF THE INVENTION

[0007] It is desirable to prevent moisture from entering the rack housing as much as possible in order to prevent rust and other problems. Even in a case where a breathing valve is provided in the steering device as described in Patent Literature 2, for example, it is desirable to prevent water droplets caused by water vapor that has passed through the breathing valve and entered the housing, or other substance, from adhering to the detector.

[0008] The present invention was made in view of the above circumstances, and is intended to achieve at least one of the first to third objects described below. The first object is to provide a position detection device capable of positioning a detector for detecting the position of a rack shaft in close proximity to the rack shaft, while still assuring the waterproof property of the rack housing. The second object is to provide a position detection device capable of preventing, for example, water droplets caused by water vapor that has passed through the breathing valve and entered the housing, from adhering to the detector. The third object is to provide a position detection device and a detection device capable of preventing a seal member which is positioned between the inner surface of a through-hole formed in a member to be mounted, such as a rack housing, and the outer surface of a support member supporting a detector, from slipping out.

[0009] For solving the aforementioned problem, one aspect of the present invention provides position detection device for detecting a position of a rack shaft that steers a steering wheel of a vehicle by moving in an axial direction, comprising: [0010] a cylindrical rack housing that houses the rack shaft; [0011] a detector that detects the position of the rack shaft relative to the rack

housing; [0012] a support member that supports the detector relative to the rack housing; and [0013] a seal member attached to the support member, [0014] wherein the rack housing has a through-hole that penetrates between the inner and outer circumference surfaces along a radial direction of the rack shaft, [0015] wherein the support member has a support portion that is positioned in the through-hole to support the detector and a fixing portion that is fixed to the rack housing, and [0016] wherein the seal member is in elastic contact with an outer surface of the support portion and an inner surface of the through-hole.
[0017] Also, for the purpose of solving the aforementioned problem, another aspect of the present invention provides a detection device, comprising: [0018] a support member having a cylindrical portion that is accommodated in a through-hole formed through a member to be mounted; [0019] a detector housed in a housing space of the cylindrical portion and supported by the support member, [0020] a seal member interposed between an inner surface of the through-hole and an outer surface of the support member; and [0021] a sealing member attached to an end of the cylindrical portion to seal the housing space, wherein the sealing member prevents the seal member from slipping out of the cylindrical portion.
Advantageous Effects of the Invention
[0022] According to the present invention, it is possible to achieve at least one of the first to third objects described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0023] FIG. **1** is a schematic diagram of a vehicle equipped with a steer-by-wire steering device having a stroke sensor as a position detection device according to the first embodiment of the present invention.
[0024] FIG. **2** is a cross-sectional view of the steering device taken along the line A-A in FIG. **1**.
[0025] FIG. **3** is a perspective view of a detection unit, a rack shaft, and a detection target attached to a rack housing.
[0026] FIG. **4** is an exploded perspective view of the detection unit.
[0027] FIG. **5** is an exploded perspective view of the detection unit.
[0028] FIG. **6**A and FIG. **6**B are cross-sectional views taken along the lines B-B and C-C in FIG. **3** respectively.
[0029] FIG. **7** is a cross-sectional view taken along the line D-D in FIG. **3**.
[0030] FIG. **8**A is a plan view showing an example of the wiring patterns of first and third wiring layers of the substrate.
[0031] FIG. **8**B is a plan view showing an example of the wiring pattern of second and fourth wiring layers of the substrate.
[0032] FIG. **9** is a plan view of the wiring patterns of the first, second, third, and fourth wiring layers of the substrate, superimposed on one another.
[0033] FIG. **10**A and FIG. **10**B respectively show before and after laser welding a support member and a sealing member.
[0034] FIG. **11** is a cross-sectional view of the seal member in its natural state with no external force acting on it.
[0035] FIG. **12** is a cross-sectional view of a hub unit equipped with a detection device according to a second embodiment.
[0036] FIG. **13**A is a cross-sectional view of the detection device and its periphery.
[0037] FIG. **13**B is an enlarged partial view of FIG. **13**A.
[0038] FIG. **14** is a perspective view showing the end of the cylindrical portion in the support member and the sealing member.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

[0039] FIG. **1** is a schematic diagram of a vehicle equipped with a steer-by-wire steering device **1** having a stroke sensor **10** as a position detection device according to the first embodiment of the present invention.

[0040] As shown in FIG. **1**, a steering device **1** includes a stroke sensor **10**, a rack shaft **11**, ball joints **12** provided at both ends of the rack shaft **11**, tie rods **13** connected to the rack shaft **11** via the ball joints **12**, a worm reduction mechanism **14** having a pinion gear **141** meshed with the rack teeth **111** of the rack shaft **11**, an electric motor **15** providing axial movement force to the rack shaft **11** via the worm reduction mechanism **14**, a steering wheel **16** operated by the driver, a steering angle sensor **17** detecting a steering angle of the steering wheel **16**, and a steering controller **18** that controls the electric motor **15** based on the steering angle detected by the steering angle sensor **17**. The rack shaft **11** steers the left and right steering wheels **19** (front wheels) by moving in the axial direction. The stroke sensor **10** detects the position of the rack shaft **11** in the vehicle width direction.

[0041] The rack shaft **11** is housed in a cylindrical rack housing **2** fixed to the car body and supported by a pair of rack bushings **201** located at both ends of the rack housing **2**. In FIG. **1**, the rack housing **2** is shown in two-dotted lines, and the rack shaft **11** and the worm reduction mechanism **14** inside the rack housing **2** are shown in solid lines. The rack housing **2** is equipped with a breathing valve **202** that allows air and water vapor to pass into and out of the rack housing **2**, but blocks water droplets. The worm reduction mechanism **14** has a worm wheel **142** and a worm gear **143**, and the pinion gear **141** is fixed to the worm wheel **142**. The worm gear **143** is fixed to the motor shaft **151** of the electric motor **15**.

[0042] The electric motor **15** generates torque by the motor current supplied by the steering controller **18** and rotates the worm wheel **142** and the pinion gear **141** via the worm gear **143**. As the pinion gear **141** rotates, the rack shaft **11** moves axially forward and backward over a predetermined travel range along the vehicle width direction, and the left and right steering wheels **19** are steered. The rack shaft **11** can move from the neutral position when the steering angle is zero to the right and left sides of the vehicle in the vehicle width direction. In FIG. **1**, a range R**1** in which the rack shaft **11** can move in the vehicle width direction is indicated by both arrows.

[0043] FIG. **2** shows a cross-sectional view of the steering device **1** taken along the line A-A in FIG. **1**. The stroke sensor **10** consists of a detection target (i.e., an object to be detected) **110** fixed to the rack shaft **11**, a rack housing **2**, a substrate **3** as a detector to detect the position of the rack shaft **11** relative to the rack housing **2**, a metal plate **101** arranged parallel to the substrate **3**, a support member **4** that supports the substrate **3** against the rack housing **2**, a seal member **5** attached to the support member **4**, a sealing member **6** sealing the substrate **3**, a power supply unit **71** and a calculation unit **72** (see FIG. **1**). The power supply unit **71** and the calculation unit **72** are connected to the substrate **3** by a cable **73** with a connector **731** at one end, as shown in FIG. **1**. The substrate **3**, metal plate **101**, support member **4**, seal member **5**, and sealing member **6** constitute a detection unit **100** in the stroke sensor **10**.

[0044] FIG. **3** is a perspective view of the detection unit **100**, the rack shaft **11**, and the detection target **110** attached to the rack housing **2**. FIG. **4** and FIG. **5** are exploded perspective views of the detection unit **100**. FIG. **6**A and FIG. **6**B are cross-sectional views taken along the lines B-B and C-C in FIG. **3** respectively. FIG. **7** is a cross-sectional view taken along the line D-D in FIG. **3**.

[0045] The rack housing **2** is made of die-cast aluminum alloy and has a housing hole **20** that accommodates the rack shaft **11**. The rack housing **2** also has a through-hole **21** that penetrates the rack housing between inner and outer circumferential surfaces along the radial direction of the rack shaft **11**. The through-hole **21** penetrates the rack housing **2** between an inner circumferential surface **2***a* and an outer circumferential surface **2***b* of the housing hole **20**. The outer circumferential surface **2***b* of the rack housing **2** has a portion thereof as a flat mounting surface **2***c* for mounting the support member **4**. The direction of penetration of the through-hole **21** is perpendicular to the mounting surface **2***c* and the central axis C of the rack housing **2**.

[0046] The rack shaft **11** is a shaft-shaped member made of steel, such as carbon steel, and the detection target **110** is attached to the outer circumference surface **11***a* of the rack shaft **11**, for example by welding. The detection target **110** is made of a material with a higher magnetic permeability than that of the rack shaft **11** or a material with a higher electrical conductivity than that of the rack shaft **11**. When a material with higher magnetic permeability than that of the rack shaft **11** is used as the material of the detection target **110**, it is desirable to use a magnetic material such as ferrite, which has high electrical resistance and is less likely to generate eddy currents. When a material with higher conductivity than that of the rack shaft **11** is used for the detection target **110**, a metal mainly composed of aluminum or copper, for example, can be used as the material. The detection target **110** may be formed as an integrated part of the rack shaft **11**.

[0047] The stroke sensor **10** detects the position of the rack shaft **11** by the position of the detection target **110**. The detection target **110** is mounted in a position facing the detection unit **100** over the entire range R.sub.1 in which the rack shaft **11** can move in the vehicle width direction, but in FIG. **3**, the detection target **110** which normally faces the detection unit **100** is in a displaced location for illustrative purposes. The surface **110***a* of the detection target **110**, facing the detection unit **100**, is flat. In this embodiment, the position of the detection target **110** is detected by the substrate **3** as a detector, using the magnetic action of the detection target **110**. Next, the composition of the substrate **3** will be described.

[0048] As shown in FIG. **2**, the substrate **3** has four layers, a first wiring layer **301**, a second wiring layer **302**, a third wiring layer **303**, and a fourth wiring layer **304** in this order from the front surface **3***a* on the detection target **110** side to the back surface **3***b*. A substrate material **30** made of a dielectric such as FR4 (glass fiber soaked into epoxy resin and thermoset) is placed between the first wiring layer **301** and the second wiring layer **302**, between the second wiring layer **302** and the third wiring layer **303**, and between the third wiring layer **303** and the fourth wiring layer **304**. The first wiring layer **301** and the fourth wiring layer **304**, which are the outer layers, are covered with a resist film **300** having electrical insulation properties. Wiring patterns are configured in the first wiring layer **301**, the second wiring layer **302**, the third wiring layer **303**, and the fourth wiring layer **304**, respectively, and these wiring patterns of the layers are connected by a via **35** at multiple locations on the substrate **3**.

[0049] FIG. **8**A is a plan view showing an example of the wiring patterns of the first wiring layer **301** and the third wiring layer **303** of the substrate **3**. FIG. **8**B is a plan view showing an example of the wiring patterns of the second wiring layer **302** and the fourth wiring layer **304** of the substrate **3**. FIG. **9** is a plan view showing the wiring patterns of the first wiring layer **301**, the second wiring layer **302**, the third wiring layer **303**, and the fourth wiring layer **304** of the substrate **3**, superimposed on one another. In FIG. **8**A, FIG. **8**B, and FIG. **9**, the wiring patterns of the third wiring layer **303** and the fourth wiring layer **304** are shown in light colors. In FIG. **8**A, FIG. **8**B, and FIG. **9**, the dimensions in the width direction (vertical direction in the drawing) relative to the dimensions in the longitudinal direction (horizontal direction in the drawing) of the substrate **3** are shown enlarged.

[0050] The substrate **3** has an excitation coil **31** that generates a magnetic field, and a first detection coil **32** and a second detection coil **33** that detect the magnetic field generated by the excitation coil **31**. The excitation coil **31** is formed in the first wiring layer **301** and the third wiring layer **303** along the perimeter of the substrate **3**. The first detection coil **32** is formed in the first wiring layer **301** and the third wiring layer **303**. The second detection coil **33** is formed in the second wiring layer **302** and the fourth wiring layer **304**. At one end of the substrate **3** in the longitudinal direction, a through-hole group **34** comprising a plurality of through-holes **341** to **346** is provided to be connected to a connector terminal **102** (see FIG. **4** and FIG. **5**) for connecting the power supply unit **71** and the calculation unit **72** by a cable **73** and a connector **731** (see FIG. **1**).

[0051] The first detection coil **32** comprises a pair of curved portions **321**, **322** whose shape as viewed from the front surface **3***a* side of the substrate **3** has a sine waveform shape. The second detection coil **33** comprises a pair of curved portions **331**, **332** whose shape as viewed from the front surface **3***a* side of the substrate **3** has a cosine waveform shape. The excitation coil **31**, the first detection coil **32**, and the second detection coil **33** are formed in a loop shape each, with their beginnings and ends connected to the through-holes **341** to **346** of the through-hole group **34**. A pair of curved portions **321**, **322** of the first detection coil **32** are connected by the via **35** at the other end in the longitudinal direction of the substrate **3**. The pair of curved portions **331**, **332** of the second detection coil **33** are connected by the via **35** at the other end in the longitudinal direction of the substrate **3**.
[0052] The excitation coil **31** generates an alternating magnetic field in a direction perpendicular to the substrate **3** by means of a high-frequency current supplied from the power supply unit **71** via the cable **73**. The magnetic flux of the AC magnetic field generated by the excitation coil **31** is also chained to the detection target **110**. The magnetic flux chained to the detection target **110** affects the intensity distribution of the magnetic flux on the substrate **3**. When the detection target **110** is made of a high permeability material, the magnetic flux is concentrated on the detection target **110**, resulting in a higher magnetic flux density in the area corresponding to the detection target **110**. If the detection target **110** is made of a high conductivity material, the eddy currents generated in the detection target **110** reduce the magnetic flux density in the area corresponding to the detection target **110**.
[0053] The spacing of the pair of curved portions **321**, **322** of the first detection coil **32** and that of the pair of curved portions **331**, **332** of the second detection coil **33** varies in the width direction of the substrate **3**. Thus, the intensity of the magnetic field detected by the first detection coil **32** and the second detection coil **33** varies according to the positions of the rack shaft **11** and the detection target **110**.
[0054] The first detection coil **32** and the second detection coil **33** output an output signal(s), which is the voltage induced by the AC magnetic field generated by the excitation coil **31**, to the calculation unit **72** via the cable **73**. The calculation unit **72** calculates the position of the detection target **110** based on the output signal(s) and transmits the calculation results to the steering controller **18**. The magnitude of the voltage induced in the first detection coil **32** and the second detection coil **33** varies within a range of less than one cycle while the rack shaft **11** moves from the moving end on one side of the axial direction to the moving end on the other side of the axial direction. Due to the difference in the shape of the first detection coil **32** and the second detection coil **33**, the phase of magnitude change is different by 90° in the voltage induced in the first detection coil **32** and the second detection coil **33** when the rack shaft **11** moves. Therefore, the stroke sensor **10** can detect the absolute position of the rack shaft **11** over the entire range R**1** in which the rack shaft **11** is movable in the axial direction.
[0055] In addition, a plurality of mounting holes **36** for fixing the substrate **3** to the support member **4** are formed through the front surface **3***a* and the back surface **3***b*. Each of the mounting holes **36** is formed at the center in the width direction of the substrate **3** in a position that does not overlap the wiring patterns of the excitation coil **31**, the first detection coil **32**, and the second detection coil **33**.
[0056] The metal plate **101** is made of iron, for example, and is ferromagnetic and conductive. The metal plate **101** is positioned parallel to the substrate **3** to enhance the magnetic coupling between the excitation coil **31**, the first detection coil **32** and the second detection coil **33**, and to suppress the influence of external electromagnetic waves. A plurality of mounting holes **101***a* are formed in the metal plate **101**. Each of the mounting holes **101***a* penetrates through the metal plate **101** in the thickness direction.
[0057] The support member **4** is made of electrically insulating resin and has in one body a support portion **41** that supports the substrate **3** and the metal plate **101**, a fixing portion **42** that is fixed to

the rack housing **2**, and a connector holding portion **43** that holds the connector **731**. In the present embodiment, four fixing portions **42** are provided in the support member **4**, and a metal collar **103** is formed by in-mold at each of the fixing portions **42**. The fixing portions **42** are fixed to the rack housing **2** by means of bolts **104** (see FIG. **3** and FIG. **6**) inserted into the collars **103** and screwed into the bolt holes **22** formed in the rack housing **2**.

[0058] The support portion **41** is at least partially disposed in the through-hole **21** of the rack housing **2**. The seal member **5** is in clastic contact with the outer surface **41***a* of the support portion **41** and the inner surface **21***a* of the through-hole **21**. Hereinafter, the rack shaft **11** side in the through-hole **21** is referred to as an “inner side (lower side)” and the opposite side is referred to as an “outer side (upper side).” The substrate **3** is positioned at the inner side in the through-hole **21**, deeper than the metal plate **101**.

[0059] The metal plate **101** is in-mold molded into the support portion **41** of the support member **4**. Resin of the support member **4** enters the plurality of mounting holes **101***a* in the metal plate **101** during the molding of the support member **4**. An inner rib **411** is formed on the inner side of the through-hole **21** deeper than the metal plate **101** in the support portion **41**. An outer rib **412** is formed on the outer side (upper side) of the through-hole **21** higher than the metal plate **101** in the support portion **41**. The inner ribs **411** and the outer rib **412** contribute to weight reduction and rigidity increase of the support member **4**. The surface of the metal plate **101** facing toward the inner side of the through-hole **21** is exposed through the gap of the inner rib **411**, but the surface of the metal plate **101** facing toward the outer side (upper side) of the through-hole **21**, is covered by the resin of the support member **4** for rust prevention and waterproofing, and is not exposed to the outside.

[0060] The substrate **3** is supported by the end portion on the rack shaft **11** side in the support portion **41**. Specifically, the back surface **3***b* of the substrate **3** is in contact with the inner rib **411**, and the substrate **3** is supported by a plurality of support projections **413** protruding from the inner rib **411** to the inner side of the through-hole **21**. In the state before the substrate **3** is attached to the support member **4**, each of the support projections **413** is cylindrical as shown in FIG. **5**. When attaching the substrate **3** to the support member **4**, insert each of the plurality of support projections **413** into the plurality of mounting holes **36** of the substrate **3**, then heat the tip of the support projections **413** while pressing the portion melted by heating onto the front surface **3***a* of the substrate **3** to form a disk-shaped mooring portion **413***a* (see FIG. **6**). With this mooring portion **413***a*, the substrate **3** is stopped from slipping out of the support projection **413** and is supported by the support portion **41**. The support portion **41** has a frame portion **414** surrounding the substrate **3**.

[0061] The sealing member **6**, like the support member **4**, is made of electrically insulating resin, and the substrate **3** is hermetically sealed between the sealing member **6** and the support portion **41** of the support member **4**. In other words, a housing space **410** that houses the substrate **3** is hermetically sealed by the sealing member **6**. This deters, for example, condensation droplets from adhering to the substrate **3**, when water vapor enters the rack housing **2** through the breathing valve **202**. In the present embodiment, the support member **4** and the sealing member **6** are joined by laser welding.

[0062] The sealing member **6** has a detection target-facing surface **6***a* facing the detection target **110**, which is formed in a flat shape. A substrate-facing surface **6***b* of the scaling member **6**, which faces the substrate **3**, has a plurality of recesses **60** to prevent interference with the mooring portion **413***a* of the support projection **413**, as shown in FIG. **4** and FIG. **6**B. The sealing member **6** has an annular protrusion **61** that is positioned inside the frame portion **414** of the support member **4**. The annular protrusion **61** controls misalignment of the sealing member **6** with respect to the support member **4** when laser welding is performed. The outer portion of the annular protrusion **61** in the sealing member **6** is hereinafter referred to as an “outer edge portion **62**.”

[0063] FIG. **10**A and FIG. **10**B show the conditions before and after laser welding in the frame portion **414** of the support portion **41** in the support member **4** and in the periphery of the seal member **5**. The frame portion **414** has an edge surface **414***a* located on the inner side of the through-hole **21** deeper than the substrate **3**. Inside the frame portion **414**, the housing space **410** is formed to accommodate the substrate **3**. The support portion **41** has a protrusion **415** protruding from the edge surface **414***a* of the frame portion **414** to the further inner side of the through-hole **21**. The frame portion **414** and the protrusion **415** are formed in an annular shape to surround the substrate **3**.

[0064] When performing laser welding, as shown in FIG. **10**A, the laser beam Lr is irradiated from the detection target-facing surface **6***a* of the sealing member **6**, while pressing the outer edge portion **62** of the sealing member **6** against a tip surface **415***a* of the protrusion **415** of the support member **4**. The laser beam Lr penetrates the outer edge portion **62** of the sealing member **6** and strikes the tip surface **415***a* of the protrusion **415**, dissolving the protrusion **415** and a part of the outer edge portion **62** of the scaling member **6** in contact with the protrusion **415**. In other words, in the present embodiment, the sealing member **6** is made of light-transmissive resin that transmits the laser beam Lr, and the support member **4** is made of light-absorbing resin that absorbs the laser beam Lr.

[0065] In this laser welding, the protrusion **415** of the support member **4** and the outer edge portion **62** of the sealing member **6** are joined all around by the relative movement of the support member **4** and the sealing member **6** to the laser torch **74**. When the resin dissolved by the laser beam Lr solidifies, the support member **4** and the sealing member **6** are welded together at the solidified portions. The height of the protrusions **415** after laser welding becomes shorter than the height of the protrusions **415** before laser welding, as shown in FIG. **10**B.

[0066] In order to ensure laser welding, it is desirable that the support member **4** and sealing member **6** be made of the same type of resin. For example, if the support member **4** is made of nylon 66, it is desirable that the sealing member **6** be made of nylon 66 as well. If the support member **4** is made of PBT (polybutylene terephthalate), it is desirable that the sealing member **6** be made of PBT as well. The support member **4** can be made light absorbent, for example, by blending carbon black.

[0067] The outer surface **41***a* of the support portion **41** of the support member **4** has an elastic contact surface **41***b* with which the seal member **5** is in clastic contact, an opposing surface **41***c* that faces the inner surface **21***a* of the through-hole **21** of the rack housing **2** at a position closer to the inner surface **21***a* of the through-hole **21** than the elastic contact surface **41***b*, and a step surface **41***d* between the elastic contact surface **41***b* and the opposing surface **41***c*. The step surface **41***d* is perpendicular to the elastic contact surface **41***b* and the opposing surface **41***c* and is oriented toward the inner side of the through-hole **21**. The seal member **5** is disposed between the step surface **41***d* and the outer edge portion **62** of the sealing member **6**, and is held from falling from the support portion **41** by the outer edge portion **62** of the sealing member **6**.

[0068] FIG. **11** is a cross-sectional view of the seal member **5** alone in its natural state without any external force acting on it. The seal member **5** made of silicone rubber, for example, is attached to the outer circumference of the support portion **41** of the support member **4**, and is placed in the through-hole **21** of the rack housing **2**. The seal member **5** is annular around the outer circumference of the support portion **41** and has a plurality of inner lips **51** and a plurality of outer lips **52**. The respective tips of the plurality of inner lips **51** and the plurality of outer lips **52** are rounded. When the support portion **41** is placed in the through-hole **21** of the rack housing **2**, the plurality of inner lips **51** are in clastic contact with the elastic contact surface **41***b* of the support portion **41** and the plurality of outer lips **52** are in elastic contact with the inner surface **21***a* of the through-hole **21**.

[0069] In FIG. **11**, the dimension in the width direction of the seal member **5** in the natural state is indicated by W. The distance G**1** (see FIG. **10**B) between the step surface **41***d* of the support portion **41** and the outer edge portion **62** of the sealing member **6** after laser welding in the penetration direction of the through-hole **21** is larger than the width W of the seal member **5**.

Therefore, a gap is formed at least one of the following locations: between one side **5***a* of the seal member **5** and the step surface **41***d* of the support portion **41**, and between the other side **5***b* of the seal member **5** and the outer edge portion **62** of the sealing member **6**. This configuration prevents the seal member **5** from being compressed between the step surface **41***d* of the support portion **41** and the outer edge portion **62** of the sealing member **6**, and prevents the outer edge portion **62** of the sealing member **6** from being separated from the protrusion **415** of the support member **4** by the restoring force of the seal member **5**.

[0070] In FIG. **10**A, the height of the protrusion **415** before laser welding is shown by T**1**, and the minimum distance between the elastic contact surface **41***b* of the support portion **41** and the tip surface **415***a* of the protrusion **415** in the direction perpendicular to the penetration direction of the through-hole **21** is shown by D. In FIG. **11**, the thickness of the inner lip **51** of the seal member **5** at the position of the minimum distance D shown in FIG. **10**A from the edge **51***a* of the inner lip **51** which is located closest to the sealing member **6** of the plurality of inner lips **51** of the seal member **5** is shown by T**2**. T**2** is thicker than T**1**, which prevents the inner lip **51** from being sandwiched between the frame portion **414** of the support portion **41** and the outer edge portion **62** of the scaling member **6** when laser welding is performed.

Effect of the First Embodiment

[0071] According to the first embodiment described above, the seal member **5** prevents the waterproof property of the rack housing **2** from being compromised while the substrate **3**, which detects the position of the rack shaft **11**, is placed close to the rack shaft **11**. Also, since the sealing member **6** seals the substrate **3** between the substrate **3** and the support portion **41** of the support member **4**, it is possible to prevent water droplets, etc. caused by water vapor passing through the breathing valve **202** and entering the rack housing **2** from adhering to the substrate **3**. In addition, the sealing member **6** can prevent the seal member **5** from slipping out to the inner side of the through-hole **21**.

Second Embodiment

[0072] Next, a second embodiment according to the present invention will be described. FIG. **12** is a cross-sectional view of a hub unit **9** equipped with a detection device **8** according to the second embodiment. FIG. **13**A is a cross-sectional view of the detection device **8** and its periphery. FIG. **13**B is an enlarged partial view of FIG. **13**A. The hub unit **9** is attached to a knuckle that constitutes the suspension system of the vehicle and rotatably supports the vehicle wheel. The detection device **8** detects the rotation speed of the vehicle wheel.

[0073] The hub unit **9** consists of a hub wheel **91** having a wheel mounting flange **911**, a plurality of hub bolts **92** secured to the wheel mounting flange **911** by press fitting, a pair of inner wheels (inner rings) **93**, **94** attached to the outer circumference of the hub wheel **91**, an outer wheel (outer wheel) **95** disposed on the outer circumference of the pair of inner wheels **93**, **94**, a plurality of rolling elements **96** and seal members **97** disposed between the pair of inner wheels **93**, **94** and the outer wheel **95**, a retainer(s) **98** holding the plurality of rolling elements **96**, and a pulsar ring **99** attached to the inner wheel **93**.

[0074] The outer wheel **95** has a fixing flange **951** for fixing to the knuckle. The detection device **8** is partially inserted into a through-hole **950** formed in the outer wheel **95** and fixed to the outer wheel **95** by a bolt **90**. The outer wheel **95** is a member to be mounted to which the detection device **8** is attached. The through-hole **950** penetrates the outer wheel **95** in a radial direction and opens on the inner circumference surface **95***a* and outer circumference surface **95***b* of the outer wheel **95**.

[0075] The outer circumference surface of the pulsar ring **99** has a plurality of magnetic poles consisting of N and S poles that are alternating along the circumferential direction. The detection device **8** detects the rotation of the wheel by the change in the magnetic field caused by the rotation of the pulsar ring **99** and outputs the detection signal to the ABS (Antilock Brake System) controller, which is not shown in the figures.

[0076] The detection device **8** consists of a support member **81** having a cylindrical portion **811** that is accommodated in the through-hole **950** formed in the outer wheel **95**, a detector **82** accommodated in a housing space **80** of the cylindrical portion **811** and supported by the support member **81**, a seal member **83** interposed between an inner surface **950***a* of the through-hole **950** and an outer surface **811***a* of the cylindrical portion **811** of the support member **81**, a sealing member **84** attached to the end of the cylindrical portion **811** to seal the housing space **80**, and a cable **85** having a plurality of wires **851**, **852** connected to the detector **82**. The seal member **83** is held from falling from the cylindrical portion **811** by the sealing member **84**.
[0077] The support member **81** is a resin molded body formed by injection molding and has the cylindrical portion **811**, a flange portion **812** for fixing to the outer wheel **95**, and a cable holding portion **813** for holding the cable **85**. A metal collar **86** is insert molded into the flange portion **812**, and by inserting a bolt **90** into the collar **86** and screwing it into the bolt hole **952** formed in the outer wheel **95**, the support member **81** is fixed to the outer wheel **95**.
[0078] The detector **82** is composed of a substrate **821** on which an unshown wiring pattern is formed and a magnetic field detection element **822** mounted on the substrate **821**. The magnetic field detection element **822** detects the magnetic field of the magnetic poles of the pulsar ring **99**. The wires **851** and **852** of the cable **85** are connected to the electrodes formed on the substrate **821**, for example, by soldering, and transmit a detection signal of the magnetic field detection element **822**. The cable **85** has a sheath **853** that accommodates the wires **851**, **852**. The sheath **853** is in liquid-tight contact with the cable holding portion **813** of the support member **81**.
[0079] FIG. **14** is a perspective view showing the end of the cylindrical portion **811** of the support member **81** and the sealing member **84**. An annular protrusion **810** is provided at the end of the cylindrical portion **811**, and the sealing member **84** is welded to the protrusion **810**. The sealing member **84** is made of resin and formed in the shape of a disk whose diameter is smaller than the inner diameter of the through-hole **950**. As a method of welding the protrusion **810** of the support member **81** and the sealing member **84**, laser welding, in which a laser beam transmitted through the sealing member **84** is irradiated onto the tip surface **810***a* of the protrusion **810**, can be used as in the first embodiment. The protrusion **810** is welded to the sealing member **84** over the entire circumference. The sealing member **84** is made of light-transmissive resin, and the support member **81** is made of light-absorbing resin.
[0080] A pair of grooves **800** to support the substrate **821** are formed in the cylindrical portion **811**. The grooves **800** are formed recessed from the inner surface **80***a* of the cylindrical portion **811** and extend in the axial direction of the cylindrical portion **811**. The substrate **821** is in a rectangular shape where the long side direction is the axial direction of the cylindrical portion **811**, and both ends of the short side are accommodated in the pair of grooves **800** and are held inside the grooves **800** by the sealing member **84**.
[0081] The outer surface **811***a* of the cylindrical portion **811** has an elastic contact surface **811***b* which the seal member **83** is in elastic contact with, an opposing surface **811***c* that faces the inner surface **950***a* of the through-hole **950** at a position closer to the inner surface **950***a* of the through-hole **950** than the elastic contact surface **811***b*, and a step surface **811***d* between the elastic contact surface **811***b* and the opposing surface **811***c*. The seal member **83** has a plurality of inner lips **831** and a plurality of outer lips **832**, wherein the plurality of inner lips **831** are in elastic contact with the elastic contact surface **811***b* and the plurality of outer lips **832** are in elastic contact with the inner surface **950***a* of the through-hole **950**.
[0082] In the axial direction of the cylindrical portion **811**, the seal member **83** is positioned between the step surface **811***d* and the sealing member **84**. The distance G**2** between the step surface **811***d* and the sealing member **84** in the axial direction of the cylindrical portion **811** is larger than the width of the seal member **83** in the same direction in its natural state. This prevents the seal member **83** from being compressed between the step surface **811***d* of the cylindrical portion **811** and the sealing member **84**, and prevents the sealing member **84** from being separated from the

protrusion **810** of the support member **81** by the restoring force of the seal member **83**.

Effects of the Second Embodiment

[0083] According to the second embodiment described above, the sealing member **84** can seal the housing space **80** of the detector **82** in the support member **81**, preventing the entry of moisture, etc. into the housing space **80**. Also, the sealing member **84** can stop the seal member **83** from slipping out of the support member **81**, preventing moisture, etc. from entering the inside of the outer wheel **95**.

Summary of the Embodiments

[0084] Next, the technical concepts that can be grasped from the first and second embodiments described above will be described with the aid of reference numerals and the like used in the first and second embodiments. However, each reference numeral in the following description does not limit the components in the scope of the claims to the parts, etc. specifically shown in the embodiments.

[0085] According to the first feature, a position detection device (stroke sensor **10**) for detecting the position of a rack shaft **11** that steers a steering wheel **19** of a vehicle by moving in an axial direction, includes a cylindrical rack housing **2** that accommodates the rack shaft **11**; a detector (substrate **3**) that detects the position of the rack shaft **11** relative to the rack housing **2**; a support member **4** that supports the detector **3** relative to the rack housing **2**; and a seal member **5** mounted on the support member **4**, wherein the through-hole **21** that penetrates inner and outer surfaces in the radial direction of the rack shaft **11** is formed in the rack housing **2**, wherein the support member **4** has a support portion **41** that is disposed in the through-hole **21** and supports the detector **3** and a fixing portion **42** that is fixed to the rack housing **2**, and wherein the seal member **5** is in elastic contact with the outer surface **41***a* of the support portion **41** and the inner surface **21***a* of the through-hole **21**.

[0086] According to the second feature, in the position detection device **10** as described in the first feature, the detector **3** is supported at the end on the rack shaft **11** side of the support **41** and is further provided with a sealing member **6** that seals the detector **3** between the sealing member and the support **41**.

[0087] According to the third feature, in the position detection device **10** as described in the second feature, the support portion **41** has an elastic contact surface **41***b* which the seal member **5** is in elastic contact with, an opposing surface **41***c* that faces the inner surface **21***a* of the through-hole **21** at a position closer to the inner surface **21***a* of the through-hole **21** than the elastic contact surface **41***b*, and a step surface **41***d* between the elastic contact surface **41***b* and the opposing surface **41***c*, and the seal member **5** is disposed between the step surface **41***d* and the sealing member **6** and is prevented from slipping out from the support portion **41** by the sealing member **6**.

[0088] According to the fourth feature, in the position detection device **10** as described in the third feature, the distance G**1** in the penetration direction of the through-hole **21** between the step surface **41***d* and the sealing member **6** is larger than the width W of the seal member **5** in the same direction in its natural state.

[0089] According to the fifth feature, in the position detection device **10** as described in any one of the first to fourth features, the detector **3** having an excitation coil **31** that generates a magnetic field and detection coils **32**, **33** that detect the magnetic field generated by the excitation coil **31** is the substrate **3** that is configured so that the intensity of the magnetic field detected by the detection coils **32**, **33** changes according to the position of the rack shaft **11**, and further comprises a metal plate **101** arranged parallel to the substrate **3** supported by the support member **4**.

[0090] According to the sixth feature, a detection device **8** includes a support member **81** having a cylindrical portion **811** which is accommodated in a through-hole **950** formed through an attachment member (outer wheel **95**), a detector **82** accommodated in a housing space **80** of the cylindrical portion **811** and supported by the support member **81**, and a seal member **83** interposed between an inner surface **950***a* of the through-hole **950** and an outer surface **811***a* of the support

member **81**, and a sealing member **84** attached to the end of the cylindrical portion **811** and sealing the housing space **80**, wherein the sealing member **84** prevents the seal member **83** from slipping out of the cylindrical portion **811**.
[0091] According to the seventh feature, in the detection device **8** as described by the sixth feature, the cylindrical portion **811** has an elastic contact surface **811***b* that the seal member **83** is in elastic contact with, an opposing surface **811***c* that faces the inner surface **950***a* of the through-hole **950** at a position closer to the inner surface **950***a* of the through-hole **950** than the contact surface **811***b*, and a step surface **811***d* between the contact surface **811***b* and the opposing surface **811***c*, and the seal member **83** is disposed between the step surface **811***d* and the sealing member **84**.
[0092] According to the eighth feature, in the detection device **8** as described by the seventh feature, the distance G**2** between the step surface **811***d* and the sealing member **84** in the axial direction of the cylindrical portion **811** is larger than the width of the seal member **83** in the same direction in the natural state.
[0093] That is all for the description of the first and second embodiments of the present invention. The first and second embodiments described above do not limit the invention according to the scope of claims. It should also be noted that not all of the combinations of features described in the first and second embodiments are essential to the means for solving the problems of the invention. In addition, the invention can be implemented with appropriate modifications without departing from the scope and spirit of the invention. For example, the following modifications are possible.
[0094] In the first and second embodiments above, the case in which the sealing members **6** and **84** are welded to the support members **4** and **81** by laser welding is described, but not limited to this, these may be welded, for example, by ultrasonic welding. In the first embodiment above, the first detection coil **32** and the second detection coil **33** of the substrate **3** detect magnetic fields as detectors, and in the second embodiment above, the magnetic field detection element **822** of the detector **82** detects magnetic fields. However, the physical quantity detected by the detector is not limited to magnetic fields, but it may be, for example, temperature, pressure, or acceleration. Furthermore, in the second embodiment, the case in which the detection device **8** is applied to the hub unit **9** is described, but the application of the detection device **8** is not limited to this, but it can be applied to various devices and equipment.

Claims

1. A position detection device for detecting a position of a rack shaft that steers a steering wheel of a vehicle by moving in an axial direction, comprising: a cylindrical rack housing that houses the rack shaft; a detector that detects the position of the rack shaft relative to the rack housing; a support member that supports the detector relative to the rack housing; and a seal member attached to the support member, wherein the rack housing has a through-hole that penetrates between the inner and outer circumference surfaces along a radial direction of the rack shaft, wherein the support member has a support portion that is positioned in the through-hole to support the detector and a fixing portion that is fixed to the rack housing, and wherein the seal member is in elastic contact with an outer surface of the support portion and an inner surface of the through-hole.
2. The position detection device, according to claim 1, wherein the detector is supported at an end on a rack shaft side of the support portion, and wherein a sealing member is further provided for sealing the detector between the sealing member and the support portion.
3. The position detection device, according to claim 2, wherein the support portion has an elastic contact surface which the seal member is in elastic contact with, an opposing surface that faces the inner surface of the through-hole at a position closer to the inner surface of the through-hole than the elastic contact surface, and a step surface between the elastic contact surface and the opposing surface, wherein the seal member is placed between the step surface and the sealing member, and wherein the sealing member prevents the seal member from slipping out of the support portion.

4. The position detection device, according to claim 3, wherein a distance between the step surface and the sealing member in a penetration direction of the through-hole is larger than a width of the seal member in a same direction in its natural state.

5. The position detection device according to claim 1, wherein the detector is a substrate having an excitation coil that generates a magnetic field and a detection coil that detects the magnetic field generated by the excitation coil, which is configured so that an intensity of the magnetic field detected by the detection coil changes according to the position of the rack shaft, and wherein a metal plate is further provided in parallel to the substrate supported by the support member.

6. A detection device, comprising: a support member having a cylindrical portion that is accommodated in a through-hole formed through a member to be mounted; a detector housed in a housing space of the cylindrical portion and supported by the support member, a seal member interposed between an inner surface of the through-hole and an outer surface of the support member; and a sealing member attached to an end of the cylindrical portion to seal the housing space, wherein the sealing member prevents the seal member from slipping out of the cylindrical portion.

7. The detection device, according to claim 6, wherein the cylindrical portion has an elastic contact surface which the seal member is in elastic contact with, an opposing surface that faces the inner surface of the through-hole at a position closer to the inner surface of the through-hole than the elastic contact surface, and a step surface between the elastic contact surface and the opposing surface, and wherein the seal member is positioned between the step surface and the sealing member.

8. The detection device, according to claim 7, wherein a distance between the step surface and the sealing member in an axial direction of the cylindrical portion is larger than a width of the seal member in a same direction in its natural state.